

A frontier obstruction for stratified low-dimensional stability

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Abstract

Alberti, Arroyo and Santacesaria ask whether their stability theory for low-dimensional inverse problems can be extended beyond fixed-dimensional manifolds without boundary, motivated by degenerate polygonal parameter families where the number of parameters varies. A previous packet proves a fixed-dimensional boundary/corner extension. Here we show that the remaining variable-dimensional stratified upgrade cannot follow from per-stratum differential injectivity alone: a compact one-dimensional stratum can approach a zero-dimensional frontier stratum with measurements separated more slowly than every power of the parameter distance.

1 Source question

The source paper proves Hölder and Lipschitz stability estimates for nonlinear inverse problems $F(x) = y$ when the unknown x belongs to a compact subset of a finite-dimensional differentiable manifold M . One central hypothesis is that F is injective and that dF_x is injective on every tangent space. The conclusion includes the following open direction.

learning may be used to learn some of the parameters of the network, like those related to the manifold M which, as mentioned before, may be unknown.

- The manifolds considered in this work are without boundary and, by definition, of fixed dimension. However, there are situations where boundary points may be of interest, in order to cover, for instance, the case of degenerate polygons, for which the number of parameters vary. Extending the current theory to this generalised setting would enlarge the range of applicability of these results.
- Another key assumption of the main results is the injectivity of the differential of F . However, in some cases, the differential of F may not be injective. It would be interesting to investigate whether a workaround may be found by using higher order derivatives of F , as in certain higher order

The source also observes that its main theorems extend readily to a finite number of pairwise disjoint compact manifold pieces, possibly of different dimensions. That extension does not cover compact stratified parameter sets where a higher-dimensional piece accumulates on a lower-dimensional frontier piece, as happens in genuine degenerations.

2 The naive stratified extension

The tempting full upgrade would say that the source theorem remains true for a compact finite stratification $K = \bigcup S_j$ if:

- each stratum S_j is a finite-dimensional C^1 manifold;

- F is continuous and globally injective on K ;
- $F|_{S_j}$ is C^1 , and $d(F|_{S_j})_x$ is injective on $T_x S_j$ for every $x \in S_j$.

The example below shows that these conditions alone do not imply even a local Hölder inverse estimate.

3 Idea of the proof

Derivative injectivity is an infinitesimal, same-stratum condition. It says that F does not collapse tangent directions at a fixed point of a fixed stratum. It does not say how quickly the image of one stratum approaches the image of another stratum.

The map $t \mapsto e^{-1/t}$ on $t > 0$ is strictly increasing and has positive derivative at every point, so it passes the one-dimensional differential test. But the image distance from t to the frontier point 0 is $e^{-1/t}$, which is smaller than every positive power of t . This destroys all Hölder stability estimates across the frontier.

4 Counterexample

Theorem 1. *There exist a compact stratified set $K \subset \mathbb{R}$ and a continuous injective map $F : K \rightarrow \mathbb{R}$ such that:*

1. K is the union of one open one-dimensional stratum and zero-dimensional endpoint strata;
2. F is C^∞ on each stratum;
3. the differential of F is injective at every point of the one-dimensional stratum, while the zero-dimensional condition is vacuous;
4. for no $C > 0$ and no $\alpha > 0$ does

$$|x - y| \leq C|F(x) - F(y)|^\alpha, \quad x, y \in K,$$

hold.

Proof. Let

$$K = [0, 1] \subset \mathbb{R}, \quad S_0 = \{0\}, \quad S_1 = (0, 1), \quad S_2 = \{1\}.$$

This is a compact set decomposed into one open one-dimensional stratum and two zero-dimensional endpoint strata. Define

$$F(0) = 0, \quad F(t) = e^{-1/t} \quad (0 < t \leq 1).$$

Then F is continuous on K , strictly increasing, and hence injective. It is smooth on S_1 , and

$$F'(t) = t^{-2}e^{-1/t} > 0, \quad 0 < t < 1.$$

Thus $d(F|_{S_1})_t : T_t S_1 \simeq \mathbb{R} \rightarrow \mathbb{R}$ is injective for every $t \in S_1$. On S_0 and S_2 , the tangent space is $\{0\}$, so injectivity of the differential is vacuous.

Suppose that a Hölder stability estimate held for some $C > 0$ and $\alpha > 0$. Applying it to $x = t$ and $y = 0$ gives

$$t \leq Ce^{-\alpha/t}, \quad 0 < t \leq 1.$$

Equivalently,

$$\frac{t}{e^{-\alpha/t}} = te^{\alpha/t} \leq C.$$

But $te^{\alpha/t} \rightarrow \infty$ as $t \downarrow 0$, a contradiction. Therefore no positive Hölder inverse estimate holds on K . $\square$

5 Relation with the source theorem

The example does not contradict any theorem in the source paper. The source manifold theorem assumes one fixed-dimensional manifold, and its finite-union extension assumes pairwise disjoint compact pieces $K_p \subset M_p$. In a metric space, finitely many pairwise disjoint compact sets are separated by a positive distance. The obstruction here is exactly the missing frontier regime: the higher-dimensional stratum accumulates at a lower-dimensional stratum, so the stratum pieces are not separated.

Nor does the example contradict the fixed-dimensional boundary/corner partial result. Boundary and corner charts keep one ambient finite dimension and use closed convex coordinate models. The present example changes the local dimension at the frontier.

6 What a positive theorem would need

The counterexample isolates a necessary extra ingredient for a genuine stratified extension. Besides the source hypotheses on each stratum, one needs some quantitative compatibility between different strata. One possible form is a cross-stratum estimate: for every pair of distinct strata S_i, S_j that can approach one another in K , there should be constants C_{ij} and $\alpha_{ij} > 0$ such that

$$\|x - y\|_X \leq C_{ij} \|F(x) - F(y)\|_Y^{\alpha_{ij}}, \quad x \in S_i, y \in S_j$$

near their common frontier. Per-stratum differential injectivity supplies no such estimate by itself.

7 Novelty check

The cheap run indexes were searched for arXiv:2009.00574, the existing boundary/corner packet, “stratified frontier”, “variable-dimensional degeneracies”, and “degenerate polygons”. The only direct hit was the prior fixed-dimensional partial result for the same source paper.

Web searches on 2026-06-28 for exact or close phrases including “2009.00574 stratified Hölder stability”, “Inverse problems on low-dimensional manifolds degenerate polygons”, and “per-stratum injective differential Hölder stability stratified” located the source paper but no separate later answer or this obstruction.

8 References

- G. S. Albierti, A. Arroyo, and M. Santacesaria, *Inverse problems on low-dimensional manifolds*, arXiv:2009.00574.